

🌐 Live Web Application: <https://voyageai-f4pd.onrender.com>
📁 GitHub Repository: <https://github.com/SanjeetMurmu126/VoyageAI>

1. Executive Summary

VoyageAI is a full-stack, AI-powered travel planner built with **Node.js, Express, and Google Gemini 2.5 Flash**. It allows users to input any destination, trip duration, budget preference, and personal interests to generate comprehensive, day-by-day travel itineraries delivered via **Server-Sent Events (SSE)** real-time streaming.

2. Key Highlights & Features

- Real-Time AI Streaming:** Delivers instant markdown-formatted itinerary updates chunk-by-chunk using Server-Sent Events (SSE) without long loading delays.
- Precision Personalization:** Tailors daily itineraries based on budget tier (Budget-friendly, Moderate, Luxury) and custom interest preferences (e.g. food, museums, shopping, nightlife).
- Interactive Day-by-Day Dashboard:** Renders structured day blocks with morning/afternoon/evening schedules, interactive packing checkboxes, and local travel tips.
- One-Click Export & Printing:** Integrated print/PDF engine formats custom itineraries into clean, styled documents.
- Modern Dashboard Design:** Engineered with a Scandinavian Slate & Emerald minimalist aesthetic—strictly NO gradients and NO glassmorphism for maximum legibility and contrast.

3. Technical Stack & Architecture

Layer	Technology	Details
Live Hosting	Render Cloud Platform	Deployed live at https://voyageai-f4pd.onrender.com
Backend Engine	Node.js / Express.js	Manages REST endpoints, static file serving, and SSE streaming pipeline

Layer	Technology	Details
Generative AI	Google Gemini 2.5 Flash	High-speed generative AI model configured for structured streaming output
Frontend UI	HTML5 / Vanilla CSS / ES6+ JS	Dashboard interface built with Space Grotesk & IBM Plex fonts, and Lucide Icons
Streaming Protocol	Server-Sent Events (SSE)	HTTP streaming protocol for progressive chunk rendering in client browser

4. API Endpoint Reference

POST `/api/generate-itinerary`

Request Payload (JSON):

```
{
  "destination": "Kyoto, Japan",
  "days": 3,
  "budget": "Moderate",
  "interests": "historic temples, traditional tea houses, local ramen spots"
}
```

5. Local Development & Deployment

- Clone repository: `git clone https://github.com/SanjeetMurmu126/VoyageAI.git`
- Install dependencies: `npm install`
- Configure environment file `.env` with `GEMINI_API_KEY=YOUR_KEY`
- Start server: `npm start` (Open `http://localhost:3000` or visit `https://voyageai-f4pd.onrender.com`)

6. Metadata & Credits

- Project Name:** VoyageAI
- Author / Owner:** Sanjeet Murmu (`SanjeetMurmu126`)
- Live Application:** `https://voyageai-f4pd.onrender.com`
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